

Quantum Time Synchronization in Star Networks

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Abstract *We demonstrate scalable multi-user clock synchronization using time-correlated photon pairs generated via spontaneous parametric downconversion in a star network. Pairwise offsets are extracted from second-order correlations, while triangle closure provides a consistency metric. Measurements show sub-200 ps agreement, with Kalman tracking of offset and skew. ©2026 The Author(s)*

Introduction

Precise time synchronization is essential for modern communication networks, enabling applications such as distributed sensing and coordinated wireless systems. Conventional protocols such as NTP and PTP provide millisecond to sub-microsecond precision, while extensions such as White Rabbit achieve sub-nanosecond performance [1]. Emerging access network applications, including 5G and future 6G fronthaul based on C-RAN architectures, impose increasingly stringent synchronization requirements, with nanosecond-level jitter constraints [2]. Quantum time synchronization approaches aim to improve both precision [3, 4] and security, offering intrinsic protection against spoofing through the no-cloning theorem.

Many quantum synchronization works have studied free-space or satellite-based systems [5, 6], while fiber-based implementations are typically limited to point-to-point links [7, 8]. Many proposed multi-user quantum synchronization schemes often require complex entangled states or sophisticated experimental setups [9–11]. An alternative approach exploits the strong temporal correlations of photon pairs generated through spontaneous parametric downconversion (SPDC), allowing users to estimate relative clock offsets via second-order correlation measurements $g^{(2)}$ [12–14]. Path-independent extensions of this method require independent sources at each user [8], which may limit scalability.

In this work, we extend passive SPDC-based synchronization to a scalable multi-user star network using a single central source. Photon pairs are distributed through a passive splitter, enabling any pair of users to recover relative timing offsets without a shared reference. All detection timestamps are exchanged through a central hub, allowing each user to estimate offsets and skews with respect to all others in a decentralized manner. A Kalman model is used to track offset and fractional frequency skew over time, while a triangle closure method provides a network-level consistency check between different user pairs. Using

this approach, we demonstrate synchronization consistency at the sub-200 ps level across three-node subsets of the network.

Operation Principles

Spontaneous parametric downconversion (SPDC) is widely used as a source of entangled photon pairs in quantum information experiments. In this process, a nonlinear $\chi^{(2)}$ crystal is pumped with a continuous-wave laser at 780 nm, producing photon pairs near 1560 nm under appropriate phase-matching conditions. In CW operation, pair generation is sparse, with approximately one pair generated per microsecond on average.

Although sparsely distributed in time, the photons within each pair are strongly correlated in their arrival times [15]. This property enables their use as a timing signal, as proposed in [12]. Remote users detect photons and estimate relative clock offsets by computing the second-order correlation function $g^{(2)}$ [16], identifying coincident detection events corresponding to the same photon pair.

To extend this approach beyond two users, we passively split the photon pairs across a star network such that, at any given time, only two users receive photons from a given pair. This configuration enables pairwise timing measurements between all users using a single central source. The randomness of photon distribution and photons' appearance in pairs provides inherent noise filtering and increases resistance to spoofing, since an adversary cannot predict which users will receive correlated photons or when pairs will arrive.

By repeatedly estimating pairwise timing offsets over time, users can also estimate relative clock skew. We employ a Kalman model to track offset and fractional frequency skew, enabling decentralized synchronization across the network. This approach provides a robust alternative to classical protocols that rely on trusted timing references, mitigating vulnerabilities such as grandmaster attacks [17].

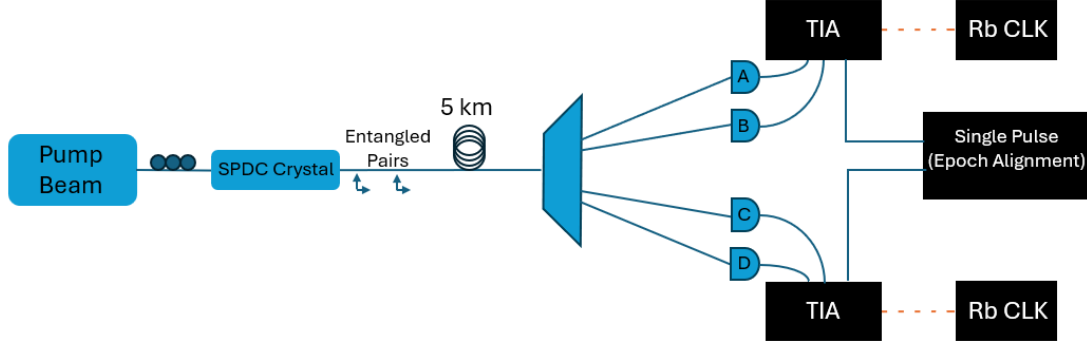


Fig. 1: Experimental setup for multi-user time synchronization using a passive SPDC source in a star network configuration. Photon pairs generated at the central node are distributed to users over fiber links, and detection timestamps are exchanged to compute pairwise correlations and estimate relative clock offsets. A single pulse that carries no timing information is used to start data acquisition.

Experimental Setup

Our experimental setup is shown in Fig. 1. We use a continuous wave (CW) laser at 780.2 nm to pump a periodically poled lithium niobate crystal held at constant temperature to produce degenerate SPDC photon pairs centered at 1560 nm. After filtering out the pump light, we send the pairs across a 5 km feeder fiber to a passive 1x8 splitter.

The pairs are split and with probability $1 - 1/N$, travel to different users. The users are equipped with off-the-shelf single photon detectors and time taggers of different brands. In this experiment, we have 4 users sharing two time taggers. The time taggers use either atomic clocks or GPS disciplined oven controlled crystal oscillators (GPSDO). The users do not use any property of entanglement aside from the temporally correlated nature of the photon pairs in this work.

Due to the difficulty of recording time tags at the same time, we use a trigger solely to align data acquisition between different time taggers and not for clock steering or timing transfer. This feature does not cost much generality because in the field, we assume the clocks are already coarsely aligned to GPS level (10 - 100 ns) and the network equipment being synchronized is already communicating asynchronously.

Results

Each user reports detection timestamps to a central hub, which then broadcasts them to all users. Pairwise clock offsets are obtained by estimating the second-order correlation function $g^{(2)}(\tau)$ from discrete coincidence counts,

$$g_{AB}^{(2)}(\tau_k) \simeq \frac{C_k}{R_A R_B T \Delta\tau}. \quad (1)$$

where C_k is the number of coincidence events in the k -th delay bin, $R_A = N_A/T$ and $R_B = N_B/T$ are the singles rates, T is the acquisition time, and $\Delta\tau$ is the bin width. The relative delay between channels A and B is taken as the value of τ_k at which $g_{AB}^{(2)}(\tau_k)$, equivalently C_k , is maximized.

In practice, the measured delay includes both propagation delay and clock offset. For a fixed star network, the fiber delays are static and can be independently calibrated using classical techniques, so temporal variations in the measured peak position are attributed to relative clock offsets. In the general case, the measured delay may be written as:

$$\tau_{AB} = t_{AB} + \delta_{AB} \quad (2)$$

where t_{AB} is the propagation delay and δ_{AB} is the true clock offset between users (A) and (B).

Because short acquisition windows can produce false maxima in $g^{(2)}$, we use triangle closure as a consistency check for peak selection. Triangle closure defines a network-level consistency constraint that is independent of the estimator used to track pairwise offsets. For three users (A), (B), and (C), the pairwise clock offsets must satisfy

$$\Delta_{ABC} = \delta_{AC} + \delta_{CB} - \delta_{AB} \approx 0. \quad (3)$$

Large deviations from this condition indicate inconsistent peak selections or other measurement artifacts. In this work, this constraint is used both as a validation metric and as the basis for triangle gating, where measurements that violate the closure condition beyond a statistical threshold are rejected or downweighted prior to filtering. Using only raw Gaussian-fit peak centers, the RMS triangle closure is less than 200 ps. We show triangle closure for an example triangle in Fig. 2.

After extracting pairwise offsets, we use a two state Kalman model to track offset and fractional frequency skew over time. The Kalman filter is used primarily as a tracking and uncertainty-propagation model, allowing measurements with different estimated uncertainties to be weighted accordingly. In this work, triangle closure is used as the primary network-level validation metric, while the Kalman-reported standard errors are interpreted as model-conditioned uncertainties. The model offsets for select pairs are contained in Fig. 3.

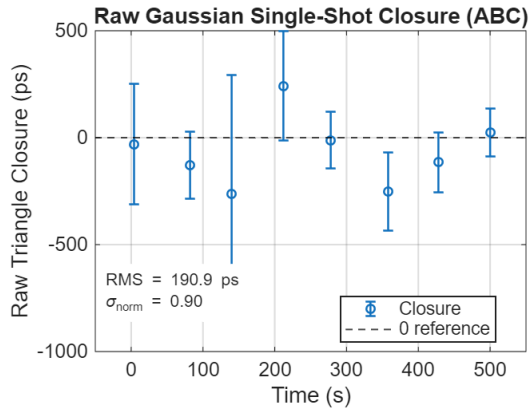


Fig. 2: Raw triangle closure residual for users A, B, and C. Error bars indicate the propagated uncertainty from the three constituent pairwise Gaussian-fit peak estimates. The residual remains centered near zero, with an RMS value of approximately 190 ps and normalized fluctuations close to unit variance, indicating good internal consistency of the raw pairwise timing measurements.

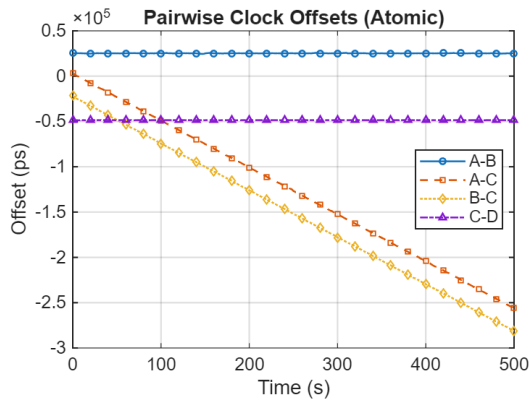
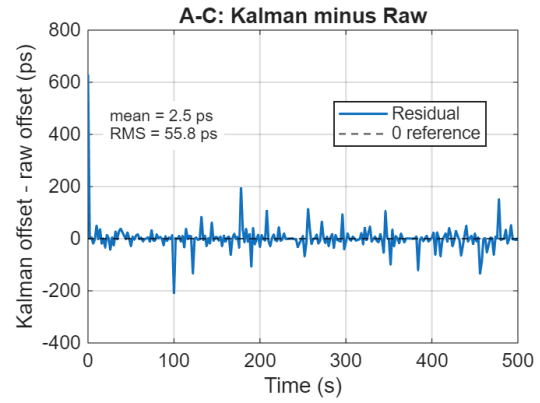


Fig. 3: Kalman-filtered pairwise clock offsets versus time for representative channel pairs in the atomic-clock configuration. The model provides smooth estimates of offset and skew, enabling tracking of relative clock evolution over the measurement interval.

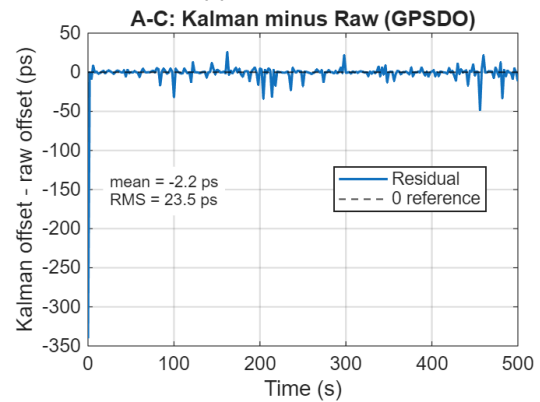
We compare the Kalman estimates with the raw centers in Fig. 4. In our Kalman model we assume constant skew, thus it is better suited to atomic clocks than GPS disciplined oscillators (GPSDO), which steer the skew using satellite signals. The atomic-clock data exhibit larger Kalman-minus-raw residuals, while the GPSDO data show closer short-term agreement between the estimator and measurements despite additional unmodeled steering dynamics.

For the atomic-clock data, raw triangle closure residuals are typically sub-200 ps, indicating good internal consistency across the network. The Kalman-filtered offsets remain broadly consistent with the raw pairwise measurements while providing smooth offset trajectories and skew estimates.

Overall, these results indicate that the present system is limited primarily by measurement noise and occasional false peak identification rather than by the Kalman model itself. Triangle closure therefore provides a more conservative and directly observable measure of synchronization consistency,



(a) Atomic clocks



(b) GPSDO

Fig. 4: Difference between the Kalman-filtered offset and the raw Gaussian-fit peak for pair A-C in the atomic (a) and GPSDO (b) configurations. The GPSDO case exhibits smaller short-term residuals, indicating closer agreement between the estimator and measurements over the observation window, while the atomic-clock data show larger deviations consistent with increased stochastic drift.

while the Kalman filter is most useful for tracking offset and skew over time. In the GPSDO case, additional dynamics associated with GPS steering are not explicitly included in the present model.

Conclusion

We demonstrated multi-user time synchronization in a star network using a passive SPDC source, enabling pairwise clock offset and skew estimation without a shared reference. Triangle closure was used as a network-level consistency check, yielding sub-200 ps residuals and indicating that performance is primarily limited by measurement noise and $g^{(2)}$ peak ambiguity. These results show that Kalman filtering is useful for tracking offset and skew over time, but its uncertainties depend on model assumptions and do not capture unmodeled dynamics such as GPS steering. Triangle closure therefore provides a direct and model-independent measure of network synchronization and establishes a robust and scalable benchmark for multi-user quantum timing.

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